

ASINGO PAUL

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PROFESSIONAL SUMMARY

Software engineer and technical lead with experience delivering web applications, backend services, cloud infrastructure, and production deployments. Experienced with Rust, Python, FastAPI, TypeScript, React, Next.js, PostgreSQL, AWS, Docker, Kubernetes, Terraform, Linux, and GitHub Actions. Hands-on experience leading development work, designing APIs and service architectures, automating deployments, and troubleshooting production systems.

TECHNICAL SKILLS

Languages: Rust, Python, TypeScript, JavaScript, SQL, Bash

Backend: Rust (Axum, Tokio, SQLx), FastAPI, SQLAlchemy, REST APIs, Microservices

Frontend & Mobile: React, Next.js, React Native (Expo)

Cloud: AWS (EC2, VPC, IAM, S3, CloudWatch, Route 53), VPS

Infrastructure: Docker, Docker Compose, Kubernetes, EKS, Terraform, Nginx

CI/CD: GitHub Actions, Jenkins, ECR, automated deployment pipelines

Systems: Linux, Ubuntu, DNS, SSL/TLS, networking, firewalls

Databases: PostgreSQL, MySQL, Redis, Amazon RDS

Tools: Git, Cloudflare, CloudWatch, Bash

PROFESSIONAL EXPERIENCE

Technical Lead / Lead Software Engineer

Specialized Engineering — Nairobi, Kenya | 2025 – Present

- Lead technical delivery from requirements and system design through development, testing, deployment, and production support.
- Design backend services with FastAPI and frontend applications with Next.js and React.
- Lead development activities including technical direction, code reviews, delivery coordination, and production support.
- Build and maintain Docker and Kubernetes deployment environments and GitHub Actions CI/CD pipelines.
- Manage Linux application environments, troubleshoot production issues, and maintain deployment workflows.
- Work with management, clients, and stakeholders to clarify requirements and demonstrate delivered functionality.

Infrastructure Engineer (Freelance / Contract)

Cloudora Solutions — Remote | Feb 2026 – Jun 2026

- Managed Linux VPS infrastructure supporting multiple production services.
- Configured Nginx reverse proxy routing, DNS, SSL certificates, and application deployments.
- Diagnosed dependency, runtime, networking, and deployment problems across production systems.
- Standardized deployment procedures and documentation to reduce manual operational errors.
- Configured SPF, DKIM, and DMARC for production email infrastructure.

Cloud & DevOps Engineer (Internship / Project-Based)

HNG — Remote | Sep 2025 – Dec 2025

- Provisioned AWS infrastructure with Terraform, including VPC, compute, IAM, and storage resources.
- Deployed containerized workloads to Kubernetes/EKS with ingress and autoscaling configurations.
- Built GitHub Actions pipelines integrating AWS ECR with EKS/ECS deployments.
- Automated AWS operations using Lambda, Systems Manager, and EventBridge.
- Troubleshot networking, IAM, container, and Kubernetes issues and documented operational procedures.
- Used CloudWatch logs and dashboards for application and infrastructure monitoring.

SELECTED PROJECTS

Velora — Company Management System

Lead developer of a company-wide operations platform with a web admin panel and a mobile app for field staff. Built modules for sales (leads, quotations, invoices, sales orders, after-sales), products and inventory (catalog, stock, stocktakes), procurement (suppliers, purchase orders, goods receipts, supplier payments), finance (general ledger, payroll, reimbursements, reports), HR, attendance, and project tracking. Implemented GPS- and camera-verified clock-ins and field updates, M-Pesa payments, and a Jenkins pipeline that builds the Android app and redeploys the Docker Compose stack.

Technologies: FastAPI, PostgreSQL, SQLAlchemy, Redis, React, TypeScript, React Native (Expo), M-Pesa, Docker, Jenkins

FileHub — Enterprise Document Management System

Technical lead for an internal document management platform. Designed the system and database, built backend APIs, developed frontend functionality, implemented role-based access control, department/sub-department switching, QR-code document verification, and managed production deployment.

Technologies: FastAPI, Next.js, PostgreSQL, Docker, GitHub Actions, AWS

DevDeploy — VPS Deployment Platform (Rust, in development)

Deployment platform that takes applications from a Git repository to a user's own VPS. Backend is a 10-crate Rust workspace: Axum REST API, PostgreSQL with SQLx migrations, SSH server provisioning, repository stack detection and Dockerfile generation, AES-256-GCM encrypted environment variables, and Argon2/JWT/TOTP/OAuth authentication.

Technologies: Rust, Axum, Tokio, SQLx, PostgreSQL, Redis, russh, Next.js

Enterprise Cloud-Native Platform

Designed and deployed a platform consisting of 16 containerized microservices. Built backend services, frontend applications, Docker configurations, and automated deployment workflows.

Technologies: FastAPI, Next.js, Docker, GitHub Actions, Linux

OpsBox — DevOps CLI Tool

Open-source command-line tool for Linux server diagnostics. Provides server health checks, CPU/memory/disk monitoring, Nginx log analysis, JSON reporting, and threshold-based alerts.

Technologies: Python, Linux, CLI, Automation

EDUCATION

Bachelor of Science in Cloud Computing and Information Security — Machakos University, 2021–2025

CERTIFICATIONS & PROGRAMS

- AWS Cloud Practitioner Essentials
- AWS AI & ML Scholars Program
- AWS Certified Solutions Architect – Associate — In Progress